

MH101 Midterm Past Paper File:03

Youtube: Star Academy

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Question No: 1 (Marks: 1) - Please choose one

_____ The set $\{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is know as set of

Natural numbers

Integers

Whole numbers

None of these

Question No: 2 (Marks: 1) - Please choose one

_____ The

$$h(x) = \frac{1}{(x-2)(x-4)}$$

domain of the function is

$(-\infty, 2) \cup (2, 4) \cup (4, +\infty)$

$(-\infty, 2] \cup \{2, 4\} \cup \{4, \infty)$

$(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$

All of these are incorrect

Question No: 3 (Marks: 1) - Please choose one

_____ If $\lim_{x \rightarrow a} f(x) = L$ then the inequality $(L - \epsilon) < f(x) < L + \epsilon$ holds in any subset of the interval

$(a - \delta, a) \cup (a, a + \delta)$

$(a - 1, a) \cup (a, a + 1)$

$(a - \epsilon, a) \cup (a, a + \epsilon)$

None of these

Question No: 4 (Marks: 1) - Please choose one

$L - \varepsilon < f(x) < L + \varepsilon$ Can be written as

$|f(x) - L| < \varepsilon$

$|f(x) - L| > \varepsilon$

$|f(x) - L| < \varepsilon + 1$

None of these

Question No: 5 (Marks: 1) - Please choose one

_____ If
a function satisfies the conditions

$f(c)$ is defined

$\lim_{x \rightarrow c^+} f(x)$

Exists

$\lim_{x \rightarrow c^+} f(x) = f(c)$

Then the function is said to be

Continuous at c

Continuous from left at c

Continuous from right at c

None of these

Question No: 6 (Marks: 1) - Please choose one

$\frac{d}{dx} [\sec x] = \text{-----}$

$\frac{\sin x}{1 - \sin^2 x}$

$\frac{-\sin x}{1 - \sin^2 x}$

$\frac{1}{1 - \sin^2 x}$

None of these

Question No: 7 (Marks: 1) - Please choose one

$$\log_b ac = \text{-----}$$

☒ $\log_b a + \log_b c$

☐ $\log_a b + \log_c b$

☐ $\log_{a+c} b$

☐ None of these

Question No: 8 (Marks: 1) - Please choose one

$$\log_b a^r = \text{-----}$$

☐ $a \log_b r$

☒ $r \log_b a$

☐ $b \log_a r$

☐ None of these

Question No: 9 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a,b) then f is ----- on (a,b)

☐ None of these

☐ Concave up

☒ Concave down

☐ Closed

Question No: 10 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative At x_0

☒ Minima

☐ Maxima

☐ None of these

Question No: 11 (Marks: 1) - Please choose one

A line is called a tangent line to the circle if it meets the circle at precisely

One point

Two points

Infinite points

None of these

Question No: 12 (Marks: 1) - Please choose one

The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at and radius

$(-4,1),\sqrt{6}$

$(-4,1),6$

$(-4,-1),\sqrt{6}$

None of these

Question No: 13 (Marks: 1) - Please choose one

The $\lim_{x \rightarrow a} f(x)$ where $f(x) = k$ (k is a constant) is equal to

k+2

k

k+1

kf

Question No: 14 (Marks: 1) - Please choose one

For any polynomial $P(x) = c_0 + c_1x + \dots + c_nx^n$ and any real number a

$\lim_{x \rightarrow a} P(x) = c_0 + c_1a + \dots + c_na^n =$

$P(a)$

$P(a+1)$

$P(a-1)$

$\frac{1}{P(a)}$

Question No: 15 (Marks: 1) - Please choose one

Polynomials are always Function

Continuous

Discontinuous

Question No: 16 (Marks: 1) - Please choose one

$$\frac{D}{Dx}[dh(x)] = \text{-----}$$

where d is a constant

$$dh(x)$$

$$dh'(x)$$

0

None of these

Question No: 17 (Marks: 1) - Please choose one

The graph $x = y^2$ is symmetric about

X-axis

Y-axis

Origin

None of these

Question No: 18 (Marks: 1) - Please choose one

Consider two function $f(x) = 3\sqrt{x}$ and $g(x) = \sqrt{x}$ what is true about these functions

$$f(x).g(x) = 3x$$

$$\frac{f(x)}{g(x)} = 3x$$

$$f(g(x)) = 3x$$

None of these

Question No: 19 (Marks: 1) - Please choose one

The formula $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is called with respect to x of the function f

Derivative

Slope

Tangent

None of these

Question No: 20 (Marks: 1) - Please choose one

Suppose that f and g are differentiable function of x then $\frac{d}{dx}\left(\frac{f}{g}\right)$

$$\frac{g \cdot f' - f \cdot g'}{g^2}$$

$$\frac{g \cdot f' + f \cdot g'}{g^2}$$

$$\frac{g \cdot f' - f \cdot g'}{g}$$

Question No: 21 (Marks: 2)

If

$$\frac{dy}{dx} = -\frac{3yx^2 + 1}{28y^3 + x^3}$$

then find the slope of the tangent line at the point (2, 0).

Question No: 22 (Marks: 3)

$$\text{Let } f(x) = \begin{cases} \frac{x^2 - x - 2}{x + 1} & \text{if } x \neq -1 \\ -3 & \text{if } x = -1 \end{cases}$$

At what points the function f is continuous and discontinuous? At point of discontinuity if any explain why it is discontinuous?

Question No: 23 (Marks: 5)

Differentiate w.r.t. x by chain rule $y = \sqrt{x^2 + 1}$

Question No: 24 (Marks: 10)

Evaluate the following limit.

$$\lim_{y \rightarrow -2} g(y) \text{ where, } g(y) = \begin{cases} y^2 + 5 & \text{if } y < -2 \\ 3 - 3y & \text{if } y \geq -2 \end{cases}$$

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